

HOMEWORK 3

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Problem 1. A complex number z is said to be algebraic if there are integers $a_0, \dots, a_n$, not all zero, such that

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0.$$

Prove that the set of all algebraic numbers is countable. Hint: For every positive integer N there are only finitely many equations with

$$n + |a_0| + |a_1| + \dots + |a_n| = N.$$

Proof. To prove this, we must construct the set of all algebraic numbers, A , and create a bijection $f : A \rightarrow \mathbb{N}$. Each polynomial has at most n roots by the Fundamental Theorem of Algebra. With our hint, we may sort the infinite possible polynomials into the form of $n + |a_0| + |a_1| + \dots + |a_n| = N$, where N is some fixed value. Doing so, we can see that we get a finite set for whatever N we choose, which is the same behavior displayed with polynomials. By doing so, we can create a sequence S_N that collects all possible finite sets, and we can create a union such that $A = \bigcup_{N=1}^{\infty} S_N, N \in \mathbb{N}$. Since $N \in \mathbb{N}$, then there does exist a bijection $f : A \rightarrow \mathbb{N}$ $\square$

Problem 2. Prove that there exist real numbers which are not algebraic. Is the set of all irrational numbers countable?

Proof. We were told in class that $\mathbb{R}$ is uncountable. We show that $\mathbb{R} \setminus A$ is uncountable by supposing for contradiction that it is. If it were countable, then $(\mathbb{R} \setminus A) \cup A$ should be countable as well, but this results in $\mathbb{R}$, which is a contradiction. So, since $(\mathbb{R} \setminus A)$ is uncountable, there exists real numbers which are not algebraic. The set of all irrationals are not countable, by the same argument shown above, but instead of using A , we use $\mathbb{Q}$. $\square$

Problem 3. Let E' be the set of all limit points of a set E . Prove that E' is closed. Prove that E and $\bar{E}$ have the same limit points. (Recall that $\bar{E} = E \cup E'$). Do E and E' always have the same limit points?

Proof. Let $p \in E'$. Let X be an arbitrary neighborhood of p . Since p is a limit point, there is a $q \in X$ such that $q \neq p$. Choose a neighborhood Y of q such that $p \notin Y$ but $Y \subset X$. q is a limit point, so there exists $r \in Y$ such that $r \neq q \implies r \neq p$. Thus, we have shown that an arbitrary limit point p contains another limit point, showing that E' is closed.

Let p be a limit point in $\bar{E}$. Let X be an arbitrary neighborhood. Since p is a limit point, there exists a $q \in X$ such that $q \neq p$. If $q \in E$, then finished. If $q \in E'$, then q is a limit point of E . Since $q \neq p$, let Y be a smaller neighborhood of q that doesn't contain p . Since q is a limit point of E , there exists a point $r \in Y$ such that $r \in E$ and $r \neq q$. Since $Y \subset X$, we have $r \in X$, and since $p \notin Y$, we have $r \neq p$.

Thus, in either case, X contains a point of E different from p . Therefore, E and $\bar{E}$ have the same limit points.

Let E be the sequence $1/n$, where $n \in \mathbb{N}$. We see that E has a limit point at 0, so $E' = \{0\}$. But taking the set of all limit points of E' gives us $E'' = \emptyset$. Therefore, they do not always have the same limit points. $\square$

Problem 4. Find a bijection between $(0, 1)$ and $[0, 1]$.

Proof. We want to show that there exists a bijection between $(0, 1)$ and $[0, 1]$. Both are subsets of $\mathbb{R}$, and are uncountably infinite. Let s_n be an arbitrary sequence of distinct points in $(0, 1)$. Let $s_1 \leftrightarrow 0$ and $s_2 \leftrightarrow 1$. From this, we see that $s_{n+2} \leftrightarrow s_n$ for all $n \in \mathbb{N}$. But, there may exist an x such that $x \notin s_n$, but is in $[0, 1]$. To resolve this, we simply let $x = x$ in our function. Let us define our function:

$$f(x) = \begin{cases} 0 & \text{if } x = s_1 \\ 1 & \text{if } x = s_2 \\ s_n & \text{if } x = s_{n+2}, n \in \mathbb{N} \\ x & \text{if } x \neq s_n \forall n \end{cases}$$

For Injectivity, the only inputs that give 0 and 1 are s_1 and s_2 . The other cases have already been established in our function, therefore $f(x)$ is injective. For surjectivity, let $p \in [0, 1]$. If $p = 0$, $f(x) = 0 \implies x = s_1$. If $p = 1$, then $f(x) = 1 \implies x = s_2$. If $p = s_n$, then $f(x) = s_n \implies x = s_{n+2}$. If $p = x$, then $f(x) = x \implies x = x$. Therefore, this is surjective. Thus, being both injective and surjective, this shows $f(x)$ is a bijection. $\square$